The lumen of the human gut is lined by a monolayer of epithelial cells that acts as a selectively permeable barrier, preventing the passage of harmful intraluminal foreign antigens, flora, and toxins into circulation while allowing digestion and absorption of essential dietary nutrients along with the transfer of electrolytes and water.

Proteins in the tight junctions of intestinal epithelial cells maintain barrier integrity, but barrier dysfunction occurs when these cells are damaged in the setting of infection, burns, shock, or hypoxia (low oxygen levels). The transcription factor HIF-1, a heterodimer composed of the macromolecules HIF-1α and HIF-1β, regulates the adaptive cellular response to hypoxia.

Researchers assessed the concentration of HIF-1 heterodimer components in human intestinal Caco-2 cells subjected to hypoxia/reoxygenation (H/R) *in vitro*. Caco-2 cells, a cell line originally derived from colon cells, were cultured under specific conditions to mimic the functional and morphological phenotype of wild-type enterocytes (ie, the cells lining the small intestine). These cells were prepared and grown as a monolayer on a collagen-coated membrane.

The monolayer was then cultured in hypoxic conditions and samples exposed to atmospheric oxygen levels (normoxia) for 30, 60, and 120 minutes. Protein levels were quantified using direct enzyme-linked immunosorbent assay (ELISA), in which an antibody linked to a reporter enzyme was used to bind and detect expression of the target molecule (analyte) in a sample. When the colorless substrate of the reporter enzyme was added, the enzyme generated a visible colored product that could be quantified based on color intensity.

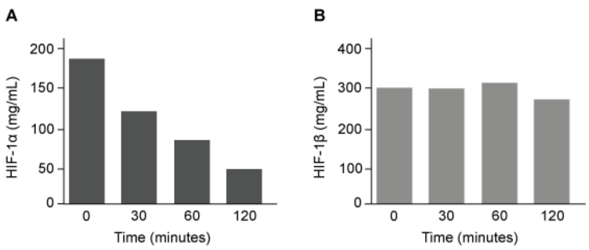

Figure 1 Concentration of (A) HIF-1α and (B) HIF-1β in Caco-2 cells subjected to H/R (Note: "0 minutes" = cells cultured in hypoxic conditions.)

Emodin, an anthraquinone compound that prevents hypoxia-induced epithelial cell disruption, was used in conjunction with a chemical HIF-1α inhibitor (HIF-1α-I) to treat Caco-2 cells in a separate experiment. Transepithelial electrical resistance (TEER) was assessed as a measure of barrier function, with higher TEER values indicating a more intact epithelial cell barrier. HIF-1α-I was found to block emodin's protective effect on epithelial barrier integrity.

Although ELISA can accurately detect the concentration of an analyte in a sample, a

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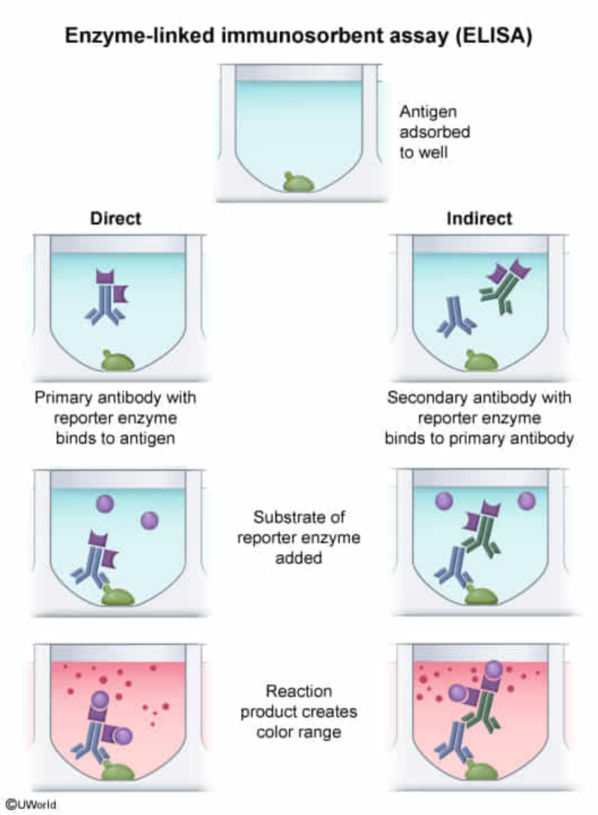
scientist claims ELISA can generate faulty results under certain experimental settings.

Inaccurate quantification of an analyte such as HIF-1α using ELISA would most likely result from any of the following EXCEPT:

- A. supplementing the sample with a denaturing agent during the initial sample preparation. [12%]
- B. adding an amount of substrate disproportionate to the amount of analyte. [16%]
- C. quantifying the color change in the presence of unbound antibodies. [18%]
- D. identifying a color change proportional to analyte concentration after adding the substrate. [52%]

Correct
52% answered correctly

Explanation:



Based on the passage, enzyme-linked immunosorbent assay (ELISA) was used to quantify HIF-1α and HIF-1β protein concentration. In ELISA, samples are first added to a 96-well microplate where the antigens (eg, proteins) become adsorbed (immobilized) to the surface of the well. A specific **primary antibody** linked to a **"reporter" enzyme** is then added to bind the antigen, as stated in the passage. The samples are washed to remove unbound antibodies, and the substrate of the reporter enzyme is added.

The enzyme reacts with its substrate to generate (ie, "report") a colored product, and if

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a color change is detected, its intensity is proportional to the amount of bound protein. Protein expression levels are ultimately quantified by comparing the color change in the well plate to the color change observed from a series of known concentration standards. **Identifying a color change** (eg, minimal, medium, intense) proportional to protein concentration would be a **normal** and **accurate finding** that indicates the **presence** and **quantity** of the **protein**.

(Choice A) Because analytes such as HIF-1α are proteins, a denaturing agent (eg, sodium dodecyl sulfate) would denature (unfold) the native protein into a linear arrangement of amino acids. This result could disassemble the antibody binding site on the protein and render ELISA *ineffective*.

(Choice B) If the amount of substrate added is disproportionate to the amount of analyte, either too much or too little substrate was added. Too much substrate can lead to excessive color change and an *inaccurate* report of higher protein levels whereas too little substrate can lead to insufficient color change and an *inaccurate* report of lower protein levels.

(Choice C) If unbound antibodies remain in the sample during the measurement of the color change, the ELISA result is *faulty* because the enzyme linked to the unbound antibody would have also created colored product. The ensuing color change would *not* be representative of actual protein levels.

Educational objective:
Enzyme-linked immunosorbent assay (ELISA) can detect and quantify proteins. Initially, a primary antibody (linked to a "reporter" enzyme) is added, which binds the antigen (protein). The samples are washed to remove unbound proteins, and the reporter enzyme substrate is added. The enzyme-substrate reaction creates a product that results in a quantifiable/detectable signal.

Time spent:0 sec
Subject:
Biology

QID:400328
System:
Digestion and Excretion

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